I Wayan Agus Darmawan

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Professional Summary

Hands-on Robotics Engineer and System Integrator experienced in designing, developing, and deploying autonomous platforms end-to-end – hardware, control loops, and the software stack around them – across quadruped, ASV, and drone systems.

Experience

PT. Ezra Robotics Teknologi Dec. 2024 – Present
Robotics Systems Lead | Robotics Engineer *Tangerang Selatan, Indonesia*

- Led system integration, hardware-software architecture, and technical deployment for autonomous quadruped robots across industrial inspection, surveillance, and public infrastructure, including LiDAR-Inertial Odometry (LIO), RTK GNSS localization, and thermal inspection.
- Executed field deployments, testing, and technical demonstrations for enterprise clients across electrical substations (PLN), oil & gas facilities (Pertamina), and agricultural plantations (PTPN).

PT. ITS Tekno Sains Sep. 2024 – Aug. 2025
Robotics Software Engineer — iCar ITS *Surabaya, Indonesia*

- Developed autonomous driving software using ROS2, Fast-LIO2, and RTK GPS for reliable navigation in GPS-available and GPS-denied environments, tuning PID controllers for vehicle stability.
- Engineered a web-based Ground Control Station using Next.js, ROSBridge, PostgreSQL, and Docker for real-time telemetry visualization, remote operation, and maintenance logging.

Barunastra ITS Roboboat Team Apr. 2022 – Nov. 2024
Electrical & Programming Division Leader | Robotics Software Engineer *Surabaya, Indonesia*

- Led the Electrical & Programming division in designing and building autonomous surface vehicles (ASVs) from the ground up, integrating LiDAR, RTK GPS, IMU, and camera sensor fusion for international competitions.
- Coordinated hardware-software integration and budget constraints across multidisciplinary teams to achieve international championship titles.

PT. ITS Tekno Sains Sep. 2023 – Dec. 2023
Robotics Software Lead — Autonomous Surface Vehicle (Project-Based) *Surabaya, Indonesia*

- Led software development for a ROS-based ASV for hydrographic bathymetry surveys, a collaborative project between PT ITS Tekno Sains, PT Robomarine Indonesia, and STTAL (end user).
- Designed autonomous navigation using Pure Pursuit with GPS RTK/IMU localization and YOLO-based buoy detection, directed field trials, and mentored MBKM ITS engineers to deliver a field-ready system.

Selected Projects

NextGen GCS | *Tauri, Go, PyTorch/TensorRT, NATS, CesiumJS* 2026

- Built a full-stack drone ground control station: live video (RTSP to WebRTC/HLS), real-time YOLO object detection, and CesiumJS 3D flight tracking, backed by a Go/NATS services layer.
- Designed a digital OpenIPC/WFB-ng video link to replace an analog FPV setup, feeding the GCS's video and inference pipeline.

Embedded Systems & SLAM | *FAST-LIO2, LIO-SAM, ESP-IDF, Jetson* 2026

- Built a Dockerized offline LiDAR-mapping pipeline (FAST-LIO2, LIO-SAM) for a Unitree Go2 EDU with Livox Mid-360, and extended the control SDK for a dual-board (Rockchip + Jetson) quadruped toward ROS2 Nav2 autonomy.
- Developed modular ESP32-C3 firmware converting PX4 AUX RC PWM into brushed-ESC speed control, with an anti-windup PID loop, encoder feedback, and a WiFi web UI for live tuning.

Technical Skills

Programming Languages: C/C++, Python, Swift, TypeScript/JavaScript, SQL
Robotics Frameworks: ROS/ROS2, ROSBridge, PID control, sensor fusion, SLAM (FAST-LIO2, LIO-SAM), Gazebo, Isaac Sim, Webots
Embedded & AI: ESP-IDF, PlatformIO, RTK GNSS, MAVLink, YOLO, PyTorch/TensorRT
Platforms & Hardware: Quadrupeds, Autonomous Surface Vehicles, Autonomous Ground Vehicles, Quadcopters/Drones

Education

Institut Teknologi Sepuluh Nopember (ITS) — B.Eng. Computer Engineering Jul. 2021 – Sep. 2025
Apple Developer Academy, Indonesia — iOS Developer Cohort Feb. 2025 – Dec. 2025